StripCheck

Checking a medicine pack against India's drug quality alerts

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Review 4: final system, experiment and report

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The problem: the list and the pack never meet

- CDSCO publishes monthly lists of batches that failed laboratory tests (NSQ) or were declared spurious.
- A patient holds a strip with a batch number on the foil. Checking means finding the month and comparing by hand.
- QR codes (top 300 brands) prove where a pack came from. They carry no test result.

7,005
alert rows, January 2019 to September 2026

1,898
rows in 2025 alone

63.5%
of flagged batches still have $\geq$12 months of shelf life at the alert

The decision: is this exact pack from a listed batch?

FLAGGED	Confirmed batch matches an NSQ entry, and product plus dates or maker agree
SUSPECTED_FAKE	Confirmed batch matches an entry listed as purported spurious
NOT_LISTED	Nothing survives exact and misread-aware search of the lists loaded. Never a certificate, never green
CANT_VERIFY	Evidence too weak after bounded steps and two questions: the agent abstains

- Advisory only: every result says "show this pack to your pharmacist". Never "safe", never "stop taking".
- Every alarm cites the list, month, failed test and source row. Only the batch is flagged, never the brand.

Why a lookup script is not enough

Reading fails confidently

- OCR read a scuffed `B` as `3` at confidence 1.0; the enhanced re-read agreed.
- An exact lookup then misses the listed batch.

Batch numbers collide

- 36 keys are shared by unrelated products.
- 2.4% of keys have under four characters.

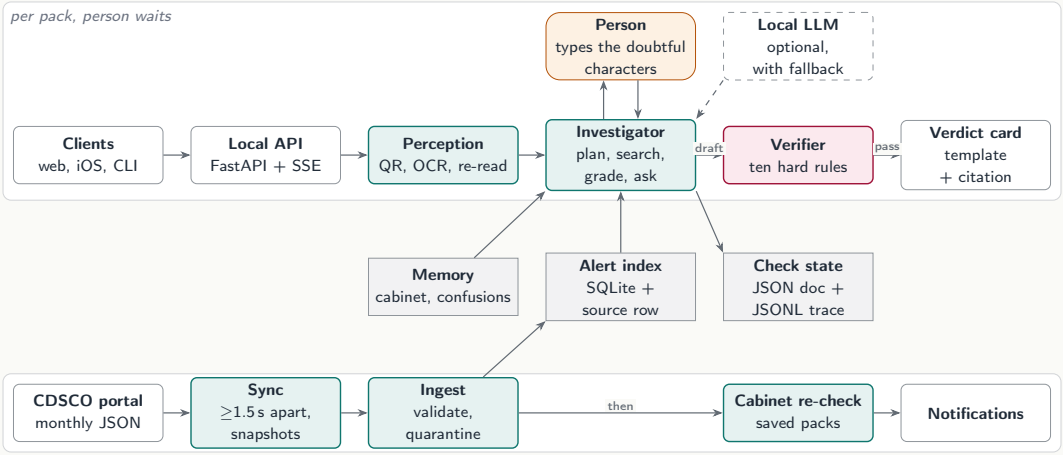
Siblings turn misreads into alarms

- 58 listed keys have a same-maker neighbour one confusable character away (`R21119` / `R21719`).
- A misread can land exactly on the sibling, and every other field agrees.

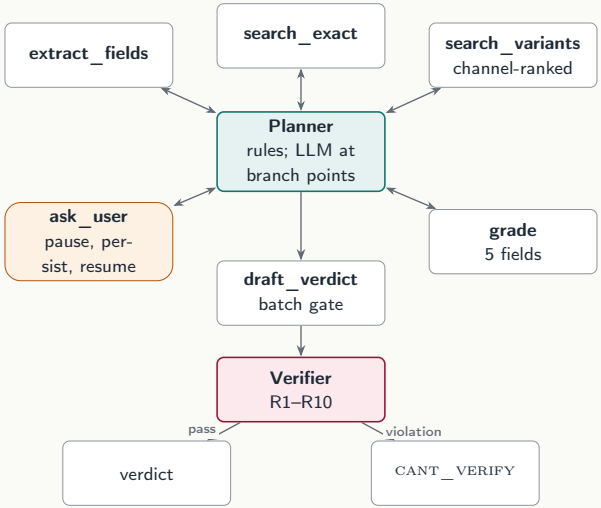
Fields do not line up

- Strips name the marketer, lists the manufacturer.
- Spurious entries list the maker as “under investigation”.

Architecture as built



The investigator loop and its bounds



Bounds in code

Steps per check	16
Questions per check	2
Substitutions per variant	2
Variant lookups	2,000
Candidates graded	20
LLM attempts (timed out)	2

Gate: no alarm unless the batch came from a QR code, the person, or two agreeing reads.

Agentic concepts and where to see them

Concept	In StripCheck	Evidence
Tool use	QR, OCR, extract, search, grade, ask, draft	one trace step per call
Agentic RAG	exact → variants → grade 5 fields → re-query or abstain	candidates and grades in the check
Planning	planner loop; hybrid LLM picks among allowed actions	*source* of each step
Observability	JSONL trace per check, live timeline	`data/traces/<check>.jsonl`
Multi-agent	supervisor, ingest, investigator, verifier	re-check report, rule results
Memory	cabinet re-check; confirmed confusions as a prior	"Prior confirmed corrections: 8 to B: 1"
HITL + guardrails	five question kinds, templates, banned phrases	question and resumed answer
Durable workflow	whole check state saved in SQLite	resume across restarts
Experimentation	lookup vs agent (rules) vs agent (hybrid)	A/B tables

Data: CDSCO's portal, ingested with provenance

- Portal JSON back to January 2019: 6,956 NSQ rows over 93 months and 49 spurious rows (snapshot of 25 October 2026).
- Every response stored with its SHA-256; every alert cites its source row.
- Imports are atomic and idempotent; bad rows are quarantined with reasons, never typed in by hand.

Quirks handled

- Latin-1 encoding, dashes arriving as "?"
- brand present in only 13.8% of rows
- multi-batch cells, GS1 prefixes, dates in batch cells
- maker and marketer in one cell
- 1.6% of rows without a manufacturing date

Evaluation set-up

Test sets

- Packs rendered from real alert rows: clean, QR, glare, misread, spurious, collision, unlisted.
- Tuned on seed 1 only; reported on seed 7.
- Eleven authored diagnostic scenarios run offline in the test suite.

Arms

- A: lookup script (QR/OCR, regex, exact hit = alarm)
- B: agent, rule planner; C: agent, hybrid planner
- each with an oracle user and with every question skipped

Metrics

- hit recall, type accuracy
- false-alarm rate, overall and on collisions
- abstention, questions per pack
- median and p95 latency
- safety violations (must be 0)

Wilson 95% intervals; exact McNemar for paired arms.

Results (oracle user)

60 rendered packs, eval seed 7, index of 6,923 alerts

Metric	Lookup oracle	Agent (rules) oracle	Agent (rules) questions skipped
Hit recall	32/40	33/40	5/40
False alarms, collisions	8/8	0/8	0/8
False alarms, siblings	4/4	0/4	0/4
False alarms, all unlisted	12/20	0/20	0/20
Abstentions	0/60	0/60	31/60
Overall accuracy	40/60	53/60	22/60
Questions per pack	0.00	0.55	0.52
Median latency (s)	1.33	1.44	1.45
Safety violations	0	0	0

A/B decision, surprises and limitations

Preregistered rule R-A
Adopt the agent if collision false alarms drop by $\geq$50% (relative) and hit recall falls by $\leq$5 points.

Decision (oracle user): adopt the agent. McNemar $p = 0.0106$.

Prediction: collisions held; clean packs did not hold; misreads did not hold.

Surprise: clean + QR recall 16/16 (lookup) vs 12/16 (agent): stricter evidence costs some true hits. With questions skipped the agent abstains instead of guessing.

Limitations

- NOT_LISTED only covers sampled, loaded lists
- all packs are synthetic; no real strip photos yet; English labels only
- the oracle answers perfectly; real people may not
- small packs per scenario; laptop CPU latency

Live demo

misread pack → one character question → flagged with citation

collision pack → not_listed where a lookup raises an alarm

trace timeline · cabinet notification

Backup: recorded trace of the misread scenario

#	Action	Observation
1	`extract_fields`	batch read as `LM24088`
2	`search_exact`	0 candidates
3	`search_variants`	1 candidate, `LM2408B`, one substitution
4	`grade`	product, dates, maker agree; batch is a variant
5	`ask_user`	character 7: printed `8` or `B`? (stored, paused)
6	`answer`	person chooses `B`; batch user-confirmed
7–9	`search, grade`	exact hit, all fields agree
10	`draft_verdict`	FLAGGED with citation; verifier passes

Backup: per-scenario counts (generated)

Correct verdicts per scenario (k/n), HITL: oracle.

Scenario	Baseline (lookup)	Agent (rules)
clean	8/8	7/8
glare	8/8	7/8
misread	0/8	6/8
qr	8/8	5/8
spurious	8/8	8/8
collision	0/8	8/8
genuine	8/8	8/8
sibling	0/4	4/4

Correct verdicts per scenario (k/n), HITL: questions skipped.

Scenario	Baseline (lookup)	Agent (rules)
clean	8/8	0/8
glare	8/8	0/8
misread	0/8	0/8
qr	8/8	5/8
spurious	8/8	0/8
collision	0/8	8/8
genuine	8/8	8/8
sibling	0/4	1/4

Backup: registered predictions and outcomes (generated)

Registered predictions and outcomes (HITL: oracle).

Prediction	Outcome	Counts
Clean packs: all arms tie	did not hold	baseline 8/8; agent_rules 7/8
Collisions: baseline false-alarms on every one; agent at most 10%	held	baseline 8/8, agent_rules 0/8
Misreads: baseline misses every flag; agent with oracle recovers at least 80% at no more than 1 question per pack	did not hold	baseline detected 0/8; agent_rules correct 6/8 at 0.75 questions per pack

Backup: verifier rules (safety)

R1	label is one of the four verdicts
R2	alarm needs accepted candidate, exact or confirmed batch, gate evidence, product match, no mismatch
R3	SUSPECTED_FAKE only for spurious entries; FLAGGED only for NSQ-type entries
R4	NOT_LISTED needs grounded batch, both searches, nothing pending, data loaded
R5	traceable citation; no banned phrase (safe, genuine, authentic, stop taking ...)
R6	at most 16 steps and 2 questions
R7	no unresolved date conflict R8 grounded provenance for every field
R9	NOT_LISTED states lists loaded and latest month R10 reviewed languages only

A draft that breaks any rule becomes CANT_VERIFY, with the rule identifiers stored. Example: an OCR-only FLAGGED draft fails R2 and R8.

Backup: deviations from the Review 2 design

Design	As built
LangGraph + checkpoints	bounded custom engine; whole check state as a JSON document in SQLite
Arize Phoenix	JSONL trace per check and live timeline; tracer optional
Streamlit	FastAPI + React web app, plus a native iOS client
PDF parsing first	portal JSON end to end; PDF helpers only
VLM second reader	not shipped; QR or two agreeing photos count as independent
Send back once	only when the LLM planner drafts too early; otherwise failing draft becomes CANT_VERIFY

Backup: team contributions

Member	Owned work	Led
Aryanraj S Singh (1BM23CD008)	investigator, verifier, planner, tracing, interfaces	R3 traces
Richik Chaudhuri (1BM23CD050)	portal sync, ingest, index, matching, cabinet, memory	data plan
Suraj B (1BM23CD061), lead	pack reading, demo packs, integration, data analysis, report	R1, demos
Sushanth Deo (1BM23CD062)	evaluation harness, test sets, A/B, results	R4 results